



Calibration based laser-induced breakdown spectroscopy (LIBS) for quantitative analysis of doped rare earth elements in phosphors

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ABSTRACT

Using laser-induced breakdown spectroscopy (LIBS), quantitative analysis of doped rare earth elements (Lanthanum (La) and Neodymium (Nd)) in phosphors has been studied and reported. Plasma is generated in air at atmospheric pressure using a high energy Q-switched Nd:YAG laser (3rd harmonic). LIBS spectra of five such samples with known composition are recorded using a high sensitive echelle spectrograph at a fixed gate delay. Using the internal standardization (IS) method, the calibration curves for La and Nd present in the phosphors are constructed. The analytical predictive capability of the LIBS system is studied through the correlation of the LIBS determined concentration of La and Nd with its known value and was found to be excellent.

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1. Introduction

Laser induced plasma (LIP) is formed when a pulsed laser is focused to a very small spot at the surface of matter, regardless of its state of matter. The condition to form the plasma is that the induced laser power density should be more than the breakdown threshold of the matter (solid, liquid, gas etc.). One can effectively utilize this plasma for excitation, ionization, sampling etc. Hence it can be used as a source for atomic emission spectroscopy. This technique can be referred as laser-induced breakdown spectroscopy (LIBS) or laser-induced plasma spectroscopy (LIPS). This is the most commonly used technique for direct determination of elements in solid samples [1–3]. Main advantage of LIBS method is its increased throughput and reduction of tedious and time consuming sample preparation procedures which can lead to contamination; ability to analyze very hard samples which are difficult to digest or dissolve; multi-elemental analysis in very short time etc. [4].

In analytical chemistry, high intense laser matter interaction has been a hot topic. The plasma produced due to this interaction, if we study spectroscopically, yield many interesting results. As time progresses in a LIP, recombination occurs and electron density decreases. Hence we get characteristic emission lines of the element present in the sample by spatially and temporally resolving this plasma. This very feature of LIBS distinguishes it

from other conventional atomic emission spectroscopy techniques [5]. Because of the very high temperature in plasma, ions as well as neutral atoms will be produced. Since the elements possess unique energy levels, it is possible to perform qualitative analysis by studying the emission spectrum of unknown samples. For quantitative analysis, the amount of an element present in the sample is determined by constructing a calibration curve of LIBS signal versus concentration by using standard samples of known concentrations i.e. the IS method. This methodology was adopted for the current study using synthesized phosphors with doped rare earth elements namely Lanthanum (La) and Neodymium (Nd) of varying concentrations.

2. Materials and methods

Synthesis: A high temperature solid state flux fusion method was used for the phosphor synthesis. The starting materials are La_2O_3 , Nd_2O_3 (Sigma Aldrich, all 99.999%), S and flux Na_2CO_3 , K_3PO_4 (Sigma Aldrich, 99.99%). Both S and Na_2CO_3 were 30–50 wt% and K_3PO_4 was 20 wt% of the total weight. The starting chemicals were thoroughly mixed using agate mortar and then heated in a muffle furnace at 1150 °C for 60 min. When the furnace was cooled down the samples were taken out and washed with distilled water 6 times and finally with a mild hydrochloric acid. The washed powder was subsequently dried and sieved.

Characterizations: X-ray powder diffraction is performed at 40 kV and 30 mA in the parallel beam method using a RIGAKU Altima IV X-ray diffractometer with $\text{Cu K}\alpha$.

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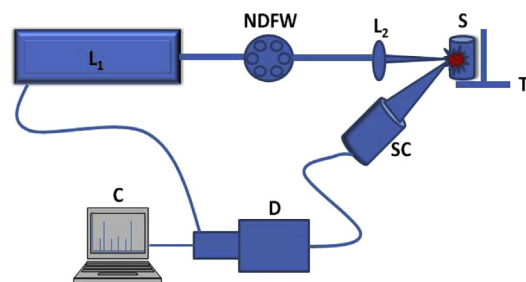
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LIBS—experimental: Fig. 1 shows experimental set up used for the current study. A 355 nm laser optical emission from Nd:YAG (Quanta Ray, Spectra Physics, USA) having pulse duration 6 ns is made to fall on a neutral density filter wheel of different optical densities to adjust the laser energy. This is then focused on to the sample using a high-energy bi-convex lens of focal length 20 cm in order to achieve appropriate breakdown threshold irradiance for different samples. The sample is placed on a motorized translation stage ensuring each laser pulse falls on a fresh sample surface. The plasma emission is collected by a light collection system which consists of UV grade quartz lenses/mirrors, and is imaged on to the spectrograph slit by using an optical fiber coupling system. A high resolution, cross dispersion echelle spectrograph-intensified charge coupled device (ICCD) system (Mechelle, Andor, Ireland) is having 195 mm focal length ($F/7$) and covers 200–975 nm spectral range in one setting with 0.05 nm wavelength resolution. With 10 μm slit, spectral resolution ($\lambda/\Delta\lambda$) will be very high with an echelle grating (52 lines/mm blazed at 575 nm). A thermoelectric cooled ICCD which is sensitive in the whole UV–vis–NIR region is used to collect the dispersed signal from spectrograph. The detector is gated in synchronization with the Nd-YAG laser pulse to get maximum signal to noise ratio. The gate delay (the time between the plasma formation and turning the ICCD on), gate width (the time during which the camera is collected the light) and the detection system were controlled by the computer and pulse delay generator which is embedded in the ICCD. An oscilloscope (TDS2024B, Tektronix, USA) and photodiode (DET10A, Thorlabs, USA) were used to monitor and calibrate the gate relative to the laser pulse. Intensity and wavelength calibrations of the spectrograph-ICCD was done using the NIST certified Deuterium–Quartz–Tungsten–Halogen [Ocean Optics, USA] and Hg–Ar lamps [Ocean Optics, USA].

3. Results and discussion

Sample composition: Compositions of the $(\text{La}_{1-x}\text{Nd}_x)_2\text{O}_2\text{S}$ series samples used for the current study is given in Table 1. Formation of single phase oxysulphide was confirmed by the XRD pattern reported in our earlier publication [6]. 3 gm of this sample was placed in a stainless steel (SS) die and pelletized using a press. The 4 mm diameter, 1 mm thick pellets were used for LIBS studies.

LIBS studies: As shown in Fig. 1, LIBS spectra were recorded from these La–Nd pellet samples using a laser line intensity of $4.4 \times 10^8 \text{ W/cm}^2$ at a detector width of 6 μs and a gate delay 700 ns. Typical LIBS spectrum of the sample is shown in Fig. 2 and the characteristic emission lines of La, Nd, O and S are marked. Four sets of such spectra are recorded in identical conditions for all the five samples. Spectral line intensities, calculated from these four spectra, are used for obtaining the calibration curves. The spectral line selection of the analyte element and the IS element is needed to be carefully made for accurate and meaningful



L₁ – Laser, NDFW – Neutral Density Filter Wheel, L₂ – Lens, S – Sample, T – Translation stages, SC – Signal Collector, D – Detector and C – Computer

Fig. 1. LIBS experimental set up used for La–Nd samples.

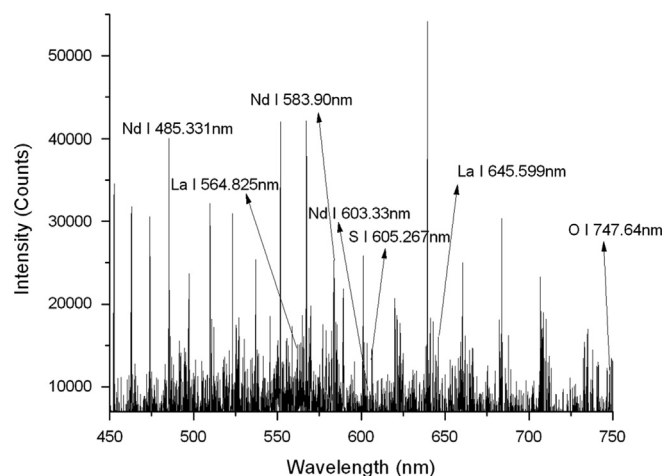


Fig. 2. LIBS spectrum of LaNdOS sample with characteristic emission lines for each element.

analytical results using the IS method. The selected lines should be strong and reasonably isolated to avoid interference with other spectral lines.

Calibration curves for La and Nd using IS method: By keeping these aspects in mind, calibration curves were constructed for Lanthanum (564.825 nm) and Neodymium (485.331 nm) by taking Oxygen (747.64 nm) as internal standard. The natural logarithm of the relative intensity ratio of the analyte (La and Nd) element and the reference (O) element lines are plotted against the natural logarithm of the given relative concentrations of the five LaNdOS samples for the calibration curves of La and Nd and depicted in Fig. 3. As evident from the figure, the calibration curves are well characterized by a straight line. The regression coefficient values of the linear fit are near unity with the best value equal to 0.994.

Analytical predictive capabilities of LIBS: In order to verify the analytical predictive capability of the LIBS system for this study the following steps have been taken. Using the linear regression equations from the constructed calibration curves of La and Nd, determined the relative concentration of La and Nd with respect to IS in five LaNdOS samples with known composition. The accuracy of the elemental determinations using the LIBS method is evaluated by their relative deviation from the certified values of these samples. Table 1 shows the correlation of the LIBS determined concentration ratio of La/O and Nd/O with its certified value and the corresponding uncertainty of the five samples using the calibration curves at 700 ns delay. In Fig. 4, the correlation of the LIBS determined concentration ratio and certified concentration ratio of La/O and Nd/O for the five

Table 1
Concentrations of La and Nd in the $(\text{La}_{1-x}\text{Nd}_x)_2\text{O}_2\text{S}$ series samples and correlation of the LIBS determined concentration ratio La/O and Nd/O with its certified values.

Sample	Concentrations (wt%)			Certified concentration ratio		LIBS determined concentration ratio (Correlation uncertainty, %)	
	La	Nd	O	La/O	Nd/O	La/O	Nd/O
1	81.26	80.24	9.3	8.738	8.628	9.209 (5.1)	9.468 (8.9)
2	64.61	65.86	9.3	6.947	7.082	6.735 (3.2)	6.920 (2.3)
3	40.00	57.81	9.3	4.302	6.216	4.165 (3.3)	5.633 (10.3)
4	23.86	41.54	9.3	2.565	4.467	2.489 (3.1)	4.507 (0.9)
5	15.86	16.77	9.3	1.705	1.803	1.776 (4.0)	1.839 (1.9)

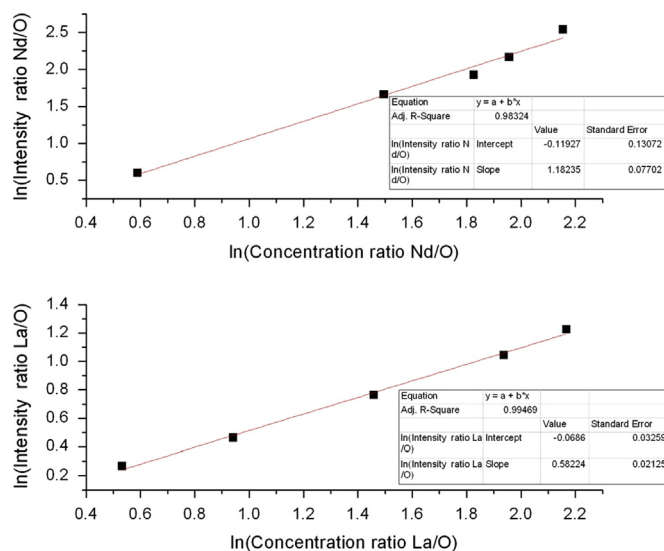


Fig. 3. Calibration curves for La and Nd at delay time 700 ns with O as IS. Linear regression equations and their coefficients are also given.

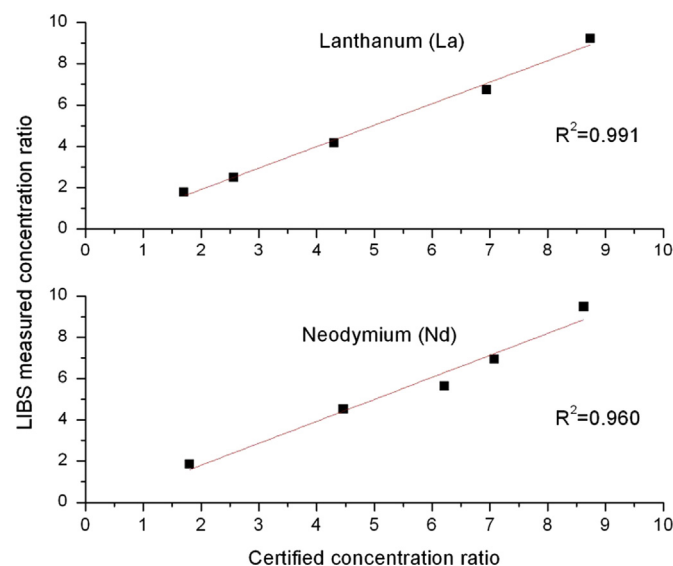


Fig. 4. Correlation between the measured LIBS concentration ratio of rare earth elements to IS and its certified concentration ratio.

samples using the calibration curve at a delay time of 700 ns are shown. It is inferred from the figure that the LIBS analytical results are fitted satisfactorily by a linear equation with the regression coefficient of 0.991 and 0.960 for La/O and Nd/O respectively. These observations underline high analytical

predictive capability of the LIBS system when a linear calibration curve is produced at an appropriate detector gate delay.

4. Conclusions

$(\text{La}_{1-x}\text{Nd}_x)_2\text{O}_2\text{S}$ series samples having five different La and Nd concentrations were prepared using high temperature solid state flux fusion method. With an ultimate aim to verify its analytical predictive capabilities, a LIBS system was assembled and the experimental conditions were optimized for these samples. LIBS spectra recorded from all five samples having varying La and Nd concentrations and calibration curves are constructed for selected spectral lines of La and Nd by choosing O as IS. The analytical performance of the LIBS system through the correlation of the LIBS determined concentration of rare earth elements with its certified value has been studied and was found to be very good. Accuracy in elemental analysis using the LIBS method is evaluated by their relative deviation from the certified values of the samples. Good agreement observed in the results of this work confirms the potential of LIBS technique for fast quantitative elemental analysis.

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